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Faculty of Science and Technology

Department of Computer Science

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PROJECT REPORT

Smartphone Cover Shop Management System

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CO2 / CO3 Assessment Criteria

Marks are intentionally left blank, as required by the submission instructions.

CO2 Criteria	0	1-2	3	4	5	Marks
Requirement fulfillment						
Validation						
Verification						

CO3 Criteria	0	1-2	3	4	5	Marks
Organization of the application						
Representation and Integration of Database						
Graphical User Interface						

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Chapter 1 — Introduction & Case Study

1.1 Case Study

The Smartphone Cover Shop Management System is a proposed three-tier marketplace desktop application designed to organize the buying and selling of smartphone covers through a centralized platform. In a traditional smartphone cover business, product information, stock levels, customer orders, sales records and discount information may be managed separately or manually. This can make it difficult for shop owners to track inventory, identify low-stock products, review sales performance and maintain accurate records. Customers may also have difficulty finding products, comparing available options and keeping track of their orders. The proposed system addresses these problems by providing a single C# Windows Forms application connected to a structured relational database.

The system has three mandatory roles. The Super Admin represents the platform owner and is responsible for controlling the overall marketplace. The Super Admin can manage shop owners, monitor platform sales and commission, manage categories, review low-rated shops and moderate customer reviews. The Admin represents a smartphone cover shop owner. Each Admin manages their own shop, products, stock and offers and can view sales and customer reviews related to their products. The Customer is the buyer who can browse available smartphone covers, search and filter products, view product details and reviews, add products to a cart, complete checkout, view orders and submit ratings and reviews.

The main entities in the system include users, shops, categories, products, carts, orders, order items, payments, reviews and offers. A user can own a shop, while a shop can contain many products. Products are organized into categories and can be added to a customer's cart. An order belongs to a customer and contains one or more products through the OrderItems junction table. Customers can make payments for orders and can write reviews for products. Products can also have active discount offers.

The platform follows a marketplace model in which customers purchase products from participating shops. The shop receives the sale amount, while the platform can calculate and monitor its commission as part of the platform-wide sales reporting process. By centralizing these activities, the system aims to reduce manual work, improve stock visibility, support accurate sales records and provide a more organized shopping experience. The design also uses role-based access so that each user reaches the appropriate dashboard and shop owners can access only their own shop data.

The three-tier model is consistent with the assignment requirement that the application connect customers with business owners through a platform operated by a Super Admin.

Chapter 2 — Functional Requirements & User Stories

2.1 Functional Requirements

2.1.1 Super Admin

1. Manage and approve/reject shop owner accounts.
2. View and manage platform users.
3. View platform-wide sales and commission information.
4. Manage product categories.
5. View low-rated shops.
6. Moderate customer reviews.

2.1.2 Admin (Shop Owner)

7. Manage the shop profile.
8. Add, update, view and delete products.
9. Monitor stock and low-stock products.
10. View sales and earnings.
11. Create and manage discount offers.
12. View product reviews and ratings.

2.1.3 Customer

13. Register and log in to the system.
14. Browse, search and filter smartphone covers.
15. View product details and reviews.
16. Add, update and remove items from the cart.
17. Complete checkout and payment.
18. View order history.
19. Write ratings and reviews.
20. View active discount offers.
21. Log out of the system.

2.2 User Stories

2.2.1 Super Admin

Manage shop owners

As a Super Admin, I can manage shop owner accounts, so that only approved shops can operate on the platform.

The Super Admin views shop owners in a data list and can approve, suspend or delete an account. The selected user's status is updated in the database.

Monitor platform sales

As a Super Admin, I can view platform sales and commission information, so that I can monitor marketplace performance.

The dashboard/report displays order totals, revenue and commission information using data from orders and shop-related records.

Manage categories

As a Super Admin, I can manage product categories, so that products can be organized consistently. The Super Admin can add, edit or delete category records. Required category information is validated before saving.

Moderate reviews

As a Super Admin, I can manage inappropriate reviews, so that the platform remains reliable.
The Super Admin reviews customer feedback and can remove an inappropriate review from the Reviews table.

2.2.2 Admin (Shop Owner)

Manage shop

As an Admin, I can manage my shop profile, so that customers see accurate shop information.
The Admin updates shop information through the shop profile form. Only the logged-in owner's shop record is modified.

Manage products

As an Admin, I can add, update and delete my products, so that my shop's catalogue remains current.
Product name, category, price and stock are validated before saving. The product is stored with the logged-in ShopID.

Monitor stock

As an Admin, I can monitor stock levels, so that I can restock products before they run out.
The inventory screen shows current stock and identifies products whose stock is below the minimum level.

View earnings

As an Admin, I can view sales and earnings, so that I can monitor my shop's performance.
The report uses order and order-item data to calculate sales for the logged-in shop only.

Manage offers

As an Admin, I can create discount offers, so that I can promote selected products.
The Admin enters the discount percentage and active dates. The offer is stored against the selected ProductID.

2.2.3 Customer

Browse products

As a Customer, I can browse smartphone covers, so that I can find products to purchase.
Available products are displayed in a list. Products with no available stock can be excluded from the shopping view.

Search and filter

As a Customer, I can search and filter products, so that I can find a suitable cover quickly.
The customer can search by keyword and use category or price filters. The results are loaded from the Products table.

Manage cart

As a Customer, I can add, update and remove cart items, so that I can prepare my purchase.
The cart stores the selected CustomerID, ProductID and quantity. Quantity is checked before the cart is updated.

Checkout

As a Customer, I can complete checkout and payment, so that my selected products become an order.
The system creates an Order and its OrderItems, calculates the total and decreases the corresponding stock.

Review products

As a Customer, I can rate and review purchased products, so that I can share my experience.
The rating must be between 1 and 5. The review is stored with the CustomerID and ProductID.

Chapter 3 — UI Navigation Diagram

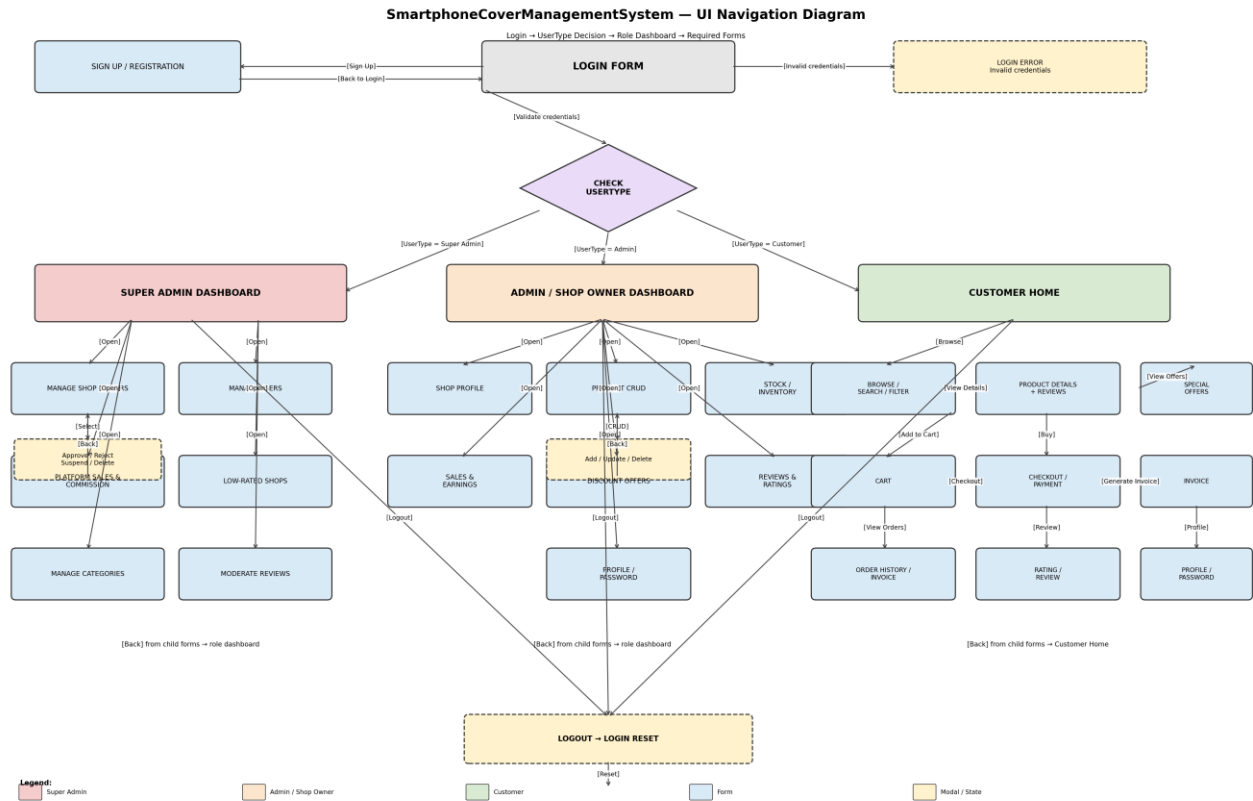


Figure 3.1: UI Navigation Diagram

The application starts at the Login form. After credentials are validated, the system checks the user's type and routes the user to the appropriate Super Admin, Admin (Shop Owner), or Customer dashboard. Each role can access only its relevant functions, while Logout returns the user to the login state.

Chapter 4 — Database Design

4.1 SQL Schema Diagram

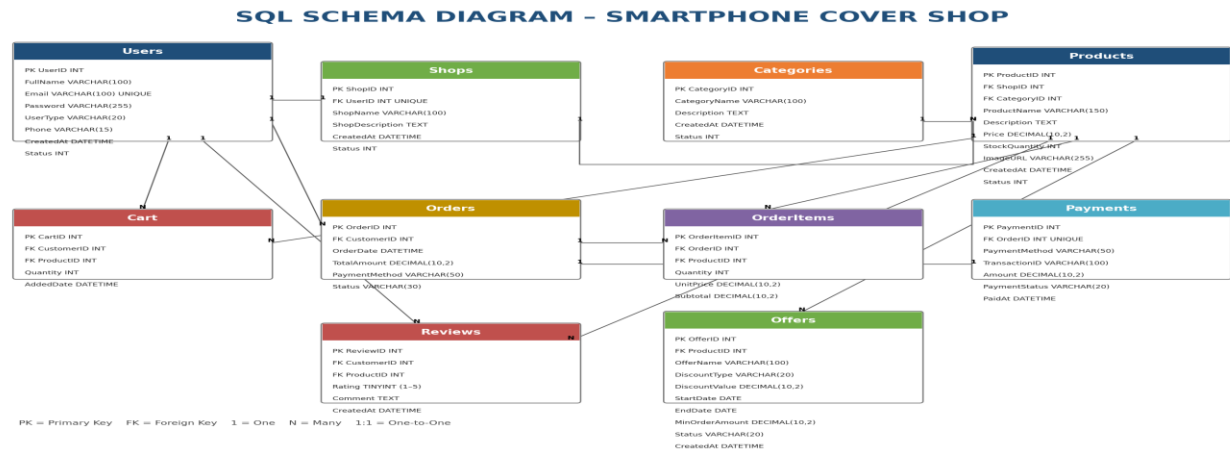


Figure 4.1: SQL Schema Diagram

4.2 Table Descriptions

4.2.1 Users

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
UserID	INT IDENTITY	PK	Unique user identifier.
FullName	VARCHAR(100)	NOT NULL	User's full name.
Email	VARCHAR(100)	UNIQUE, NOT NULL	Login email address.
Password	VARCHAR(255)	NOT NULL	User password.
UserType	VARCHAR(20)	NOT NULL	System role: super_admin, admin, or customer.
Phone	VARCHAR(15)	NULL	User contact number.
CreatedAt	DATETIME	DEFAULT	Account creation date and time.
Status	INT	NOT NULL	1 = active, 0 = inactive.

4.2.2 Shops

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
ShopID	INT IDENTITY	PK	Unique shop identifier.
UserID	INT	FK, UNIQUE	References Users.UserID; the shop owner account.
ShopName	VARCHAR(100)	NOT NULL	Shop name.
ShopDescription	TEXT	NULL	Description of the shop.
CreatedAt	DATETIME	DEFAULT	Shop creation date and time.
Status	INT	NOT NULL	1 = active, 0 = inactive.

4.2.3 Categories

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
CategoryID	INT IDENTITY	PK	Unique category identifier.
CategoryName	VARCHAR(100)	NOT NULL	Category name for smartphone covers.
Description	TEXT	NULL	Category description.
CreatedAt	DATETIME	DEFAULT	Category creation date and time.
Status	INT	NOT NULL	1 = active, 0 = inactive.

4.2.4 Products

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
ProductID	INT IDENTITY	PK	Unique product identifier.

ShopID	INT	FK, NOT NULL	References the shop that owns the product.
CategoryID	INT	FK, NOT NULL	References the product category.
ProductName	VARCHAR(150)	NOT NULL	Smartphone cover product name.
Description	TEXT	NULL	Product description.
Price	DECIMAL(10,2)	NOT NULL	Selling price of the product.
StockQuantity	INT	NOT NULL	Current available stock quantity.
ImageURL	VARCHAR(255)	NULL	Optional product image path or URL.
CreatedAt	DATETIME	DEFAULT	Product creation date and time.
Status	INT	NOT NULL	1 = active, 0 = inactive.

4.2.5 Cart

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
CartID	INT IDENTITY	PK	Unique cart-item identifier.
CustomerID	INT	FK, NOT NULL	References Users.UserID for a customer account.
ProductID	INT	FK, NOT NULL	References the selected product.
Quantity	INT	NOT NULL	Quantity selected by the customer.
AddedDate	DATETIME	DEFAULT	Date and time the item was added.

4.2.6 Orders

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
OrderID	INT IDENTITY	PK	Unique order identifier.
CustomerID	INT	FK, NOT NULL	References Users.UserID for the customer.
OrderDate	DATETIME	DEFAULT	Order creation date and time.
TotalAmount	DECIMAL(10,2)	NOT NULL	Total amount of the order.
PaymentMethod	VARCHAR(50)	NOT NULL	Selected payment method.
Status	VARCHAR(30)	NOT NULL	Order status such as Placed, Shipped or Delivered.

4.2.7 OrderItems

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
OrderItemID	INT IDENTITY	PK	Unique order-item identifier.
OrderID	INT	FK, NOT NULL	References the related order.
ProductID	INT	FK, NOT NULL	References the purchased product.
Quantity	INT	NOT NULL	Quantity purchased.
UnitPrice	DECIMAL(10,2)	NOT NULL	Product price at the time of purchase.
Subtotal	DECIMAL(10,2)	NOT NULL	Quantity multiplied by unit price.

4.2.8 Payments

Column	Data Type	Constraint	Description
Column	Data Type	Constraint	Description
PaymentID	INT IDENTITY	PK	Unique payment identifier.
OrderID	INT	FK, UNIQUE	References Orders.OrderID; one payment record per order.
PaymentMethod	VARCHAR(50)	NOT NULL	Payment method used for the order.
TransactionID	VARCHAR(100)	NULL	Optional payment transaction reference.
Amount	DECIMAL(10,2)	NOT NULL	Amount recorded for the payment.
PaymentStatus	VARCHAR(20)	NOT NULL	Payment status such as paid, failed or refunded.
PaidAt	DATETIME	DEFAULT	Payment date and time.

4.2.9 Reviews

Column	Data Type	Constraint	Description
ReviewID	INT IDENTITY	PK	Unique review identifier.
CustomerID	INT	FK, NOT NULL	References Users.UserID for the reviewer.
ProductID	INT	FK, NOT NULL	References the reviewed product.
Rating	TINYINT	CHECK 1-5	Product rating from 1 to 5.
Comment	TEXT	NULL	Optional customer review comment.
CreatedAt	DATETIME	DEFAULT	Date and time the review was submitted.

4.2.10 Offers

Column	Data Type	Constraint	Description
OfferID	INT IDENTITY	PK	Unique offer identifier.
ProductID	INT	FK, NOT NULL	References the product receiving the offer.
OfferName	VARCHAR(100)	NOT NULL	Name of the discount offer.
DiscountType	VARCHAR(20)	NOT NULL	percentage or fixed.
DiscountValue	DECIMAL(10,2)	NOT NULL	Discount amount or percentage.
StartDate	DATE	NOT NULL	Offer start date.
EndDate	DATE	NOT NULL	Offer end date.
MinOrderAmount	DECIMAL(10,2)	NULL	Optional minimum order amount.
Status	VARCHAR(20)	NOT NULL	active or inactive.
CreatedAt	DATETIME	DEFAULT	Offer creation date and time.

Note: CustomerID fields reference Users.UserID for users whose UserType is customer; no separate Customers table is used in this design.

4.3 ER Diagram

ER DIAGRAM - SMARTPHONE COVER SHOP

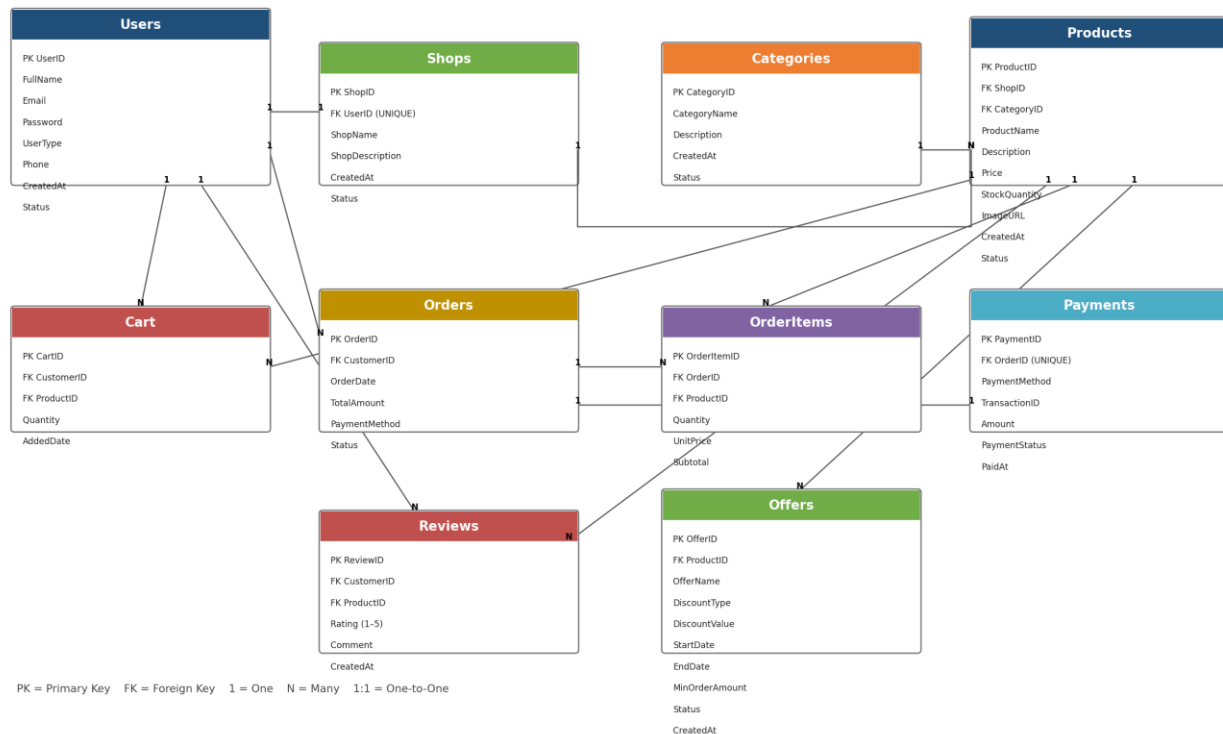


Figure 4.2: ER Diagram

4.4 Normalisation — 3NF

The database is normalized up to Third Normal Form (3NF) to reduce data redundancy and maintain data consistency.

First Normal Form (1NF)

All tables contain atomic values, with no repeating groups or multi-valued fields. Products in an order are represented through the OrderItems junction table rather than being stored as a comma-separated list.

Second Normal Form (2NF)

The database satisfies 2NF because it is in 1NF and non-key attributes depend on the complete primary key. Order-specific product information is separated into OrderItems, preventing partial dependency.

Third Normal Form (3NF)

The database satisfies 3NF because non-key attributes depend only on their table's primary key and not on other non-key attributes. User, shop, product, order, review and offer information are separated into appropriate tables. This reduces unnecessary duplication and makes inserting, updating and deleting information more consistent.

Conclusion: the Smartphone Cover Shop Management System database is designed to satisfy 1NF, 2NF and 3NF.

Chapter 5 — SQL Queries

The following feature-based queries represent the final query set used by the project design. They cover authentication, product search/filtering, cart operations, checkout, reporting, stock alerts, reviews, Super Admin monitoring and offers.

1. LOGIN — Verify credentials and role routing

```
SELECT UserID, FullName, UserType, Status
FROM Users
WHERE Email = @Email
      AND Password = @Password
      AND Status = 1;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

2. FILTER — Products by price range

```
SELECT ProductID, ProductName, Price, StockQuantity
FROM Products
WHERE Price BETWEEN @MinPrice AND @MaxPrice
      AND Status = 1;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

3. FILTER — Category and availability

```
SELECT p.ProductID, p.ProductName, p.Price, p.StockQuantity, c.CategoryName
FROM Products p
JOIN Categories c ON p.CategoryID = c.CategoryID
WHERE c.CategoryName = @Category
      AND p.StockQuantity > 0
      AND p.Status = 1;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

4. SEARCH — Products by keyword

```
SELECT ProductID, ProductName, Price, StockQuantity
FROM Products
WHERE ProductName LIKE '%' + @Keyword + '%'
      AND Status = 1;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

5. CART — Add, delete and view cart

```
INSERT INTO Cart (CustomerID, ProductID, Quantity, AddedDate)
VALUES (@CustomerID, @ProductID, @Quantity, GETDATE());

DELETE FROM Cart
WHERE CartID = @CartID;

SELECT c.CartID, p.ProductName, p.Price, c.Quantity,
      (p.Price * c.Quantity) AS LineTotal
FROM Cart c
JOIN Products p ON c.ProductID = p.ProductID
WHERE c.CustomerID = @CustomerID;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

6. CHECKOUT — Create order and order items

```
BEGIN TRANSACTION;

INSERT INTO Orders (CustomerID, OrderDate, TotalAmount, PaymentMethod, Status)
VALUES (@CustomerID, GETDATE(), @TotalAmount, @PaymentMethod, 'Placed');

DECLARE @OrderID INT = SCOPE_IDENTITY();

INSERT INTO OrderItems (OrderID, ProductID, Quantity, UnitPrice, Subtotal)
VALUES (@OrderID, @ProductID, @Quantity, @UnitPrice, @Quantity * @UnitPrice);

UPDATE Products
SET StockQuantity = StockQuantity - @Quantity
WHERE ProductID = @ProductID;

INSERT INTO Payments (OrderID, PaymentMethod, TransactionID, Amount, PaymentStatus, PaidAt)
VALUES (@OrderID, @PaymentMethod, @TransactionID, @TotalAmount, 'paid', GETDATE());

COMMIT TRANSACTION;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

7. SHOP EARNINGS — Sales grouped by shop

```
SELECT s.ShopID, s.ShopName,
       SUM(oi.Subtotal) AS TotalSales
FROM Shops s
JOIN Products p ON s.ShopID = p.ShopID
JOIN OrderItems oi ON p.ProductID = oi.ProductID
JOIN Orders o ON oi.OrderID = o.OrderID
WHERE o.Status <> 'Cancelled'
GROUP BY s.ShopID, s.ShopName;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

8. LOW STOCK ALERT

```
SELECT ProductID, ProductName, StockQuantity
FROM Products
WHERE StockQuantity <= @LowStockThreshold
AND Status = 1;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

9. REVIEWS — Reviews for a product

```
SELECT r.ReviewID, u.FullName, r.Rating,
       r.Comment, r.CreatedAt
FROM Reviews r
JOIN Users u ON r.CustomerID = u.UserID
WHERE r.ProductID = @ProductID;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

10. SUPER ADMIN — Low-rated shops

```
SELECT s.ShopID, s.ShopName,
       AVG(CAST(r.Rating AS DECIMAL(5,2))) AS AverageRating
FROM Shops s
JOIN Products p ON s.ShopID = p.ShopID
JOIN Reviews r ON p.ProductID = r.ProductID
GROUP BY s.ShopID, s.ShopName
HAVING AVG(CAST(r.Rating AS DECIMAL(5,2))) < @RatingThreshold;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

11. SUPER ADMIN — Suspend a shop owner

```
UPDATE Users
SET Status = 0
WHERE UserID = @OwnerID
AND UserType = 'admin';
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

12. OFFERS — Active offers today

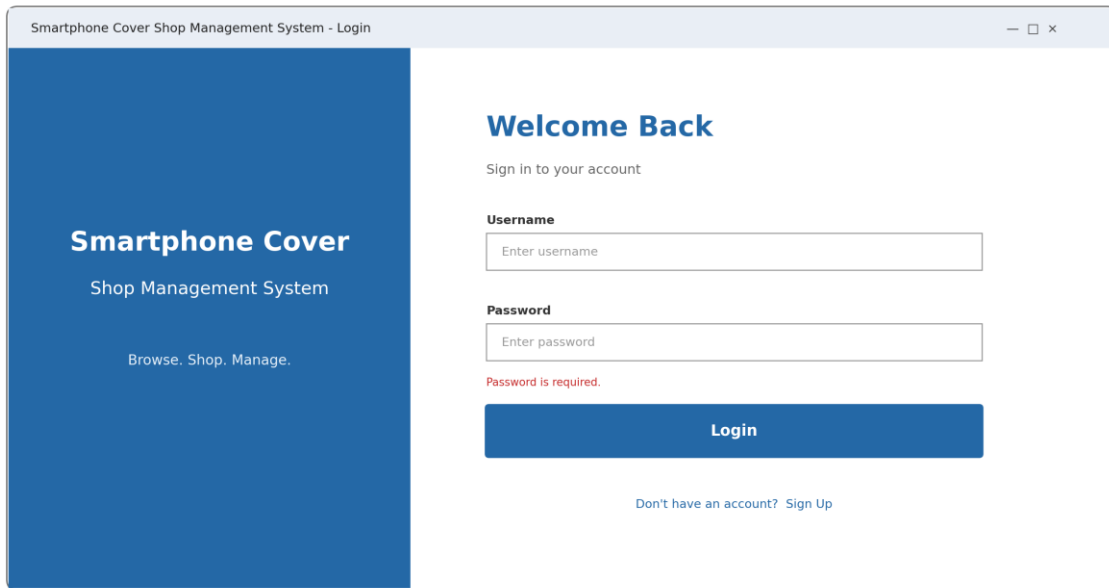
```
SELECT o.OfferID, o.OfferName, o.ProductID,
       o.DiscountType, o.DiscountValue,
       p.ProductName, p.Price
FROM Offers o
JOIN Products p ON o.ProductID = p.ProductID
WHERE o.Status = 'active'
AND CAST(GETDATE() AS DATE) BETWEEN o.StartDate AND o.EndDate;
```

Purpose: Supports the corresponding system feature and retrieves or modifies the required database records.

Chapter 6 — User Interface Design

The finalized UI designs use a simple and consistent visual style inspired by the sample project. The forms follow the approved panel sequence and maintain consistent colours, fonts, buttons and navigation.

6.1 Login



The screenshot shows a web browser window titled "Smartphone Cover Shop Management System - Login". The page is split into two main sections. On the left is a solid blue vertical panel. On the right is a white panel with the login form. The blue panel contains the text "Smartphone Cover" in large white font, "Shop Management System" in smaller white font, and "Browse. Shop. Manage." in even smaller white font. The white panel has a "Welcome Back" heading in blue, followed by the text "Sign in to your account". Below this are two input fields: "Username" with a placeholder "Enter username" and "Password" with a placeholder "Enter password". A red error message "Password is required." is displayed below the password field. A blue "Login" button is positioned below the password field. At the bottom of the white panel, there is a link that says "Don't have an account? Sign Up".

Figure 6.1: Login Form

Figure 6.1: Login Form

The Login form authenticates the user and routes the user to the appropriate role-based dashboard.

6.2 Sign Up / Registration



**SMARTPHONE COVER
SHOP MANAGEMENT SYSTEM**

★

Quality Covers. Trusted Shops.

Create Your Account

Fill in the details to register.

Full Name

User Type

Customer

▼

Username

Email

Phone Number

Password

👁

Confirm Password

👁

☐ I agree to the [Terms and Conditions](#)

Register

OR

Already have an account? [Login](#)

Figure 6.2: Sign Up / Registration Form

The registration form allows a Customer or Shop Owner to create an account and return to Login.

6.3 Super Admin Dashboard

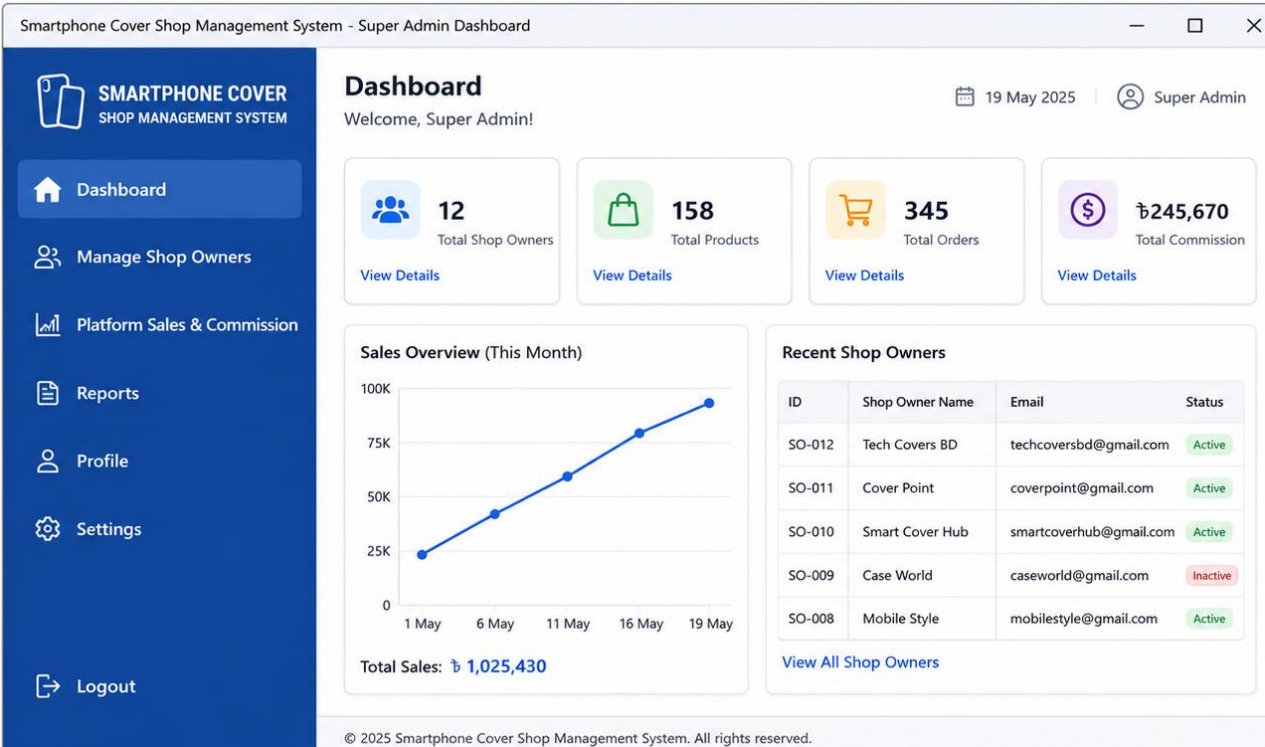


Figure 6.3: Super Admin Dashboard

The Super Admin Dashboard provides access to shop-owner management and platform-wide sales and commission functions.

6.4 Super Admin — Manage Shop Owners

This form allows the Super Admin to view, add, edit, activate/deactivate and delete Shop Owners.

Smartphone Cover Shop Management System - Manage Shop Owners

Manage Shop Owners [+ Add New Shop Owner](#)

Search by name, email or phone... All Status

ID	Shop Owner Name	Email	Phone	Join Date	Status	Action
SO-012	Tech Covers BD	techcoversbd@gmail.com	01711-223344	18 May 2025	Active	Edit Activate Delete
SO-011	Cover Point	coverpoint@gmail.com	01822-334455	15 May 2025	Active	Edit Activate Delete
SO-010	Smart Cover Hub	smartcoverhub@gmail.com	01933-445566	12 May 2025	Active	Edit Activate Delete
SO-009	Case World	caseworld@gmail.com	01678-556677	10 May 2025	Inactive	Edit Activate Delete
SO-008	Mobile Style	mobiliestyle@gmail.com	01555-667788	05 May 2025	Active	Edit Activate Delete

Showing 1 to 5 of 12 entries [« Prev](#) [1](#) [2](#) [3](#) [Next »](#)

Figure 6.4: Super Admin — Manage Shop Owners

The Super Admin can view, add, update, activate/deactivate and delete Shop Owner accounts.

6.5 Super Admin — Platform Sales & Commission Report

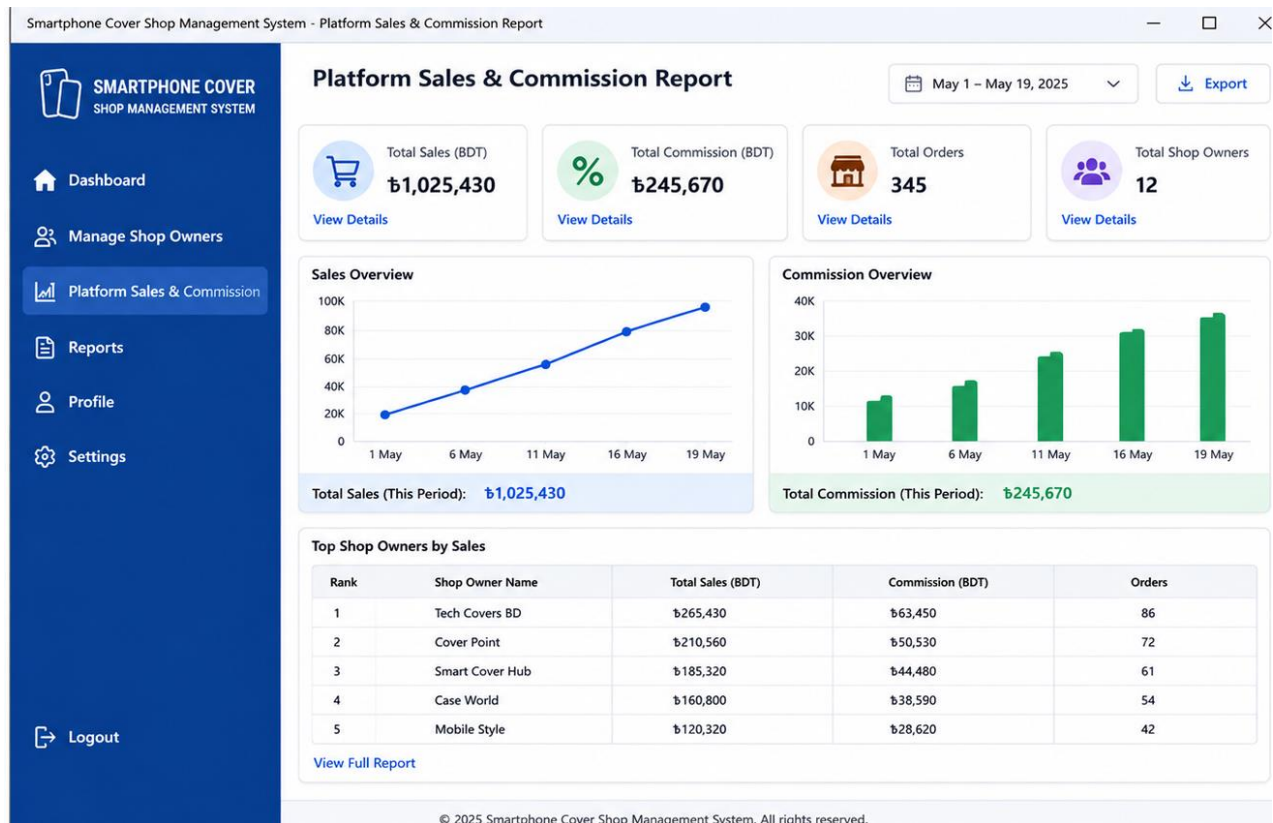


Figure 6.5: Super Admin — Platform Sales & Commission Report

The report shows overall platform sales, commission and related marketplace information within a selected date range.

6.6 Admin — Product CRUD

Smartphone Cover Shop Management System - Admin - Products

Products
Manage your shop products.

+ Add New Product Refresh

Search by product name... All Categories

ID	Product Name	Category	Price (BDT)	Stock	Status	Action
P001	 iPhone 14 Pro Max Case	iPhone	850	25	Active	 
P002	 Samsung S23 Ultra Case	Samsung	790	18	Active	 
P003	 OnePlus 11R Case	OnePlus	650	30	Active	 
P004	 iPhone 13 Leather Case	iPhone	990	12	Active	 
P005	 Samsung A54 Clear Case	Samsung	550	40	Active	 

Showing 1 to 5 of 25 products

« Prev 1 2 3 ... 5 Next »

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Figure 6.6: Admin — Product CRUD

The Shop Owner can add new products and update or delete existing products using the product DataGridView.

6.7 Admin — Stock / Inventory Dashboard

SMARTPHONE COVER
SHOP MANAGEMENT SYSTEM

Dashboard

Products

Stock / Inventory

Sales & Earnings

Orders

Profile

Settings

Logout

Smartphone Cover Shop Management System - Admin - Stock / Inventory

— □ ×

Stock / Inventory Management

View and update stock quantities.

Search by product name... 🔍

🔄 Refresh

ID	Product Name	Category	Current Stock	Update Stock	Status	Action
P001	 iPhone 14 Pro Max Case	iPhone	25	25 ⬆ ⬇ ⬆ Update	In Stock	
P002	 Samsung S23 Ultra Case	Samsung	18	18 ⬆ ⬇ ⬆ Update	In Stock	
P003	 OnePlus 11R Case	OnePlus	30	30 ⬆ ⬇ ⬆ Update	In Stock	
P004	 iPhone 13 Leather Case	iPhone	12	12 ⬆ ⬇ ⬆ Update	Low Stock	
P005	 Samsung A54 Clear Case	Samsung	40	40 ⬆ ⬇ ⬆ Update	In Stock	

Showing 1 to 5 of 25 products

« Prev 1 2 3 ... 5 Next »

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Figure 6.7: Admin — Stock / Inventory Dashboard

The Shop Owner can view and update stock quantities, with low-stock items clearly identified.

6.8 Admin — Sales & Earnings Report

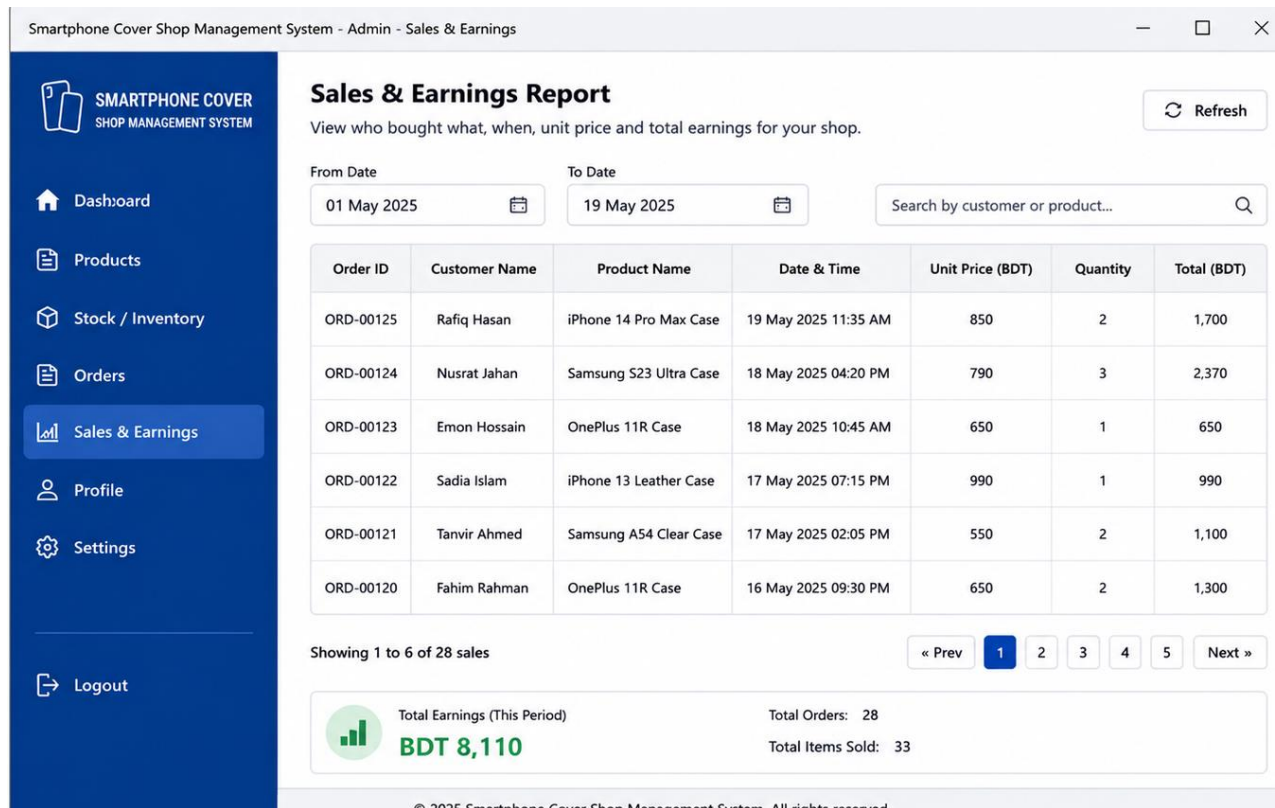


Figure 6.8: Admin — Sales & Earnings Report

The Shop Owner can view customer, product, date/time, unit price, quantity and total earnings, filtered by date range and limited to the owner's shop.

6.9 Customer — Home / Browse

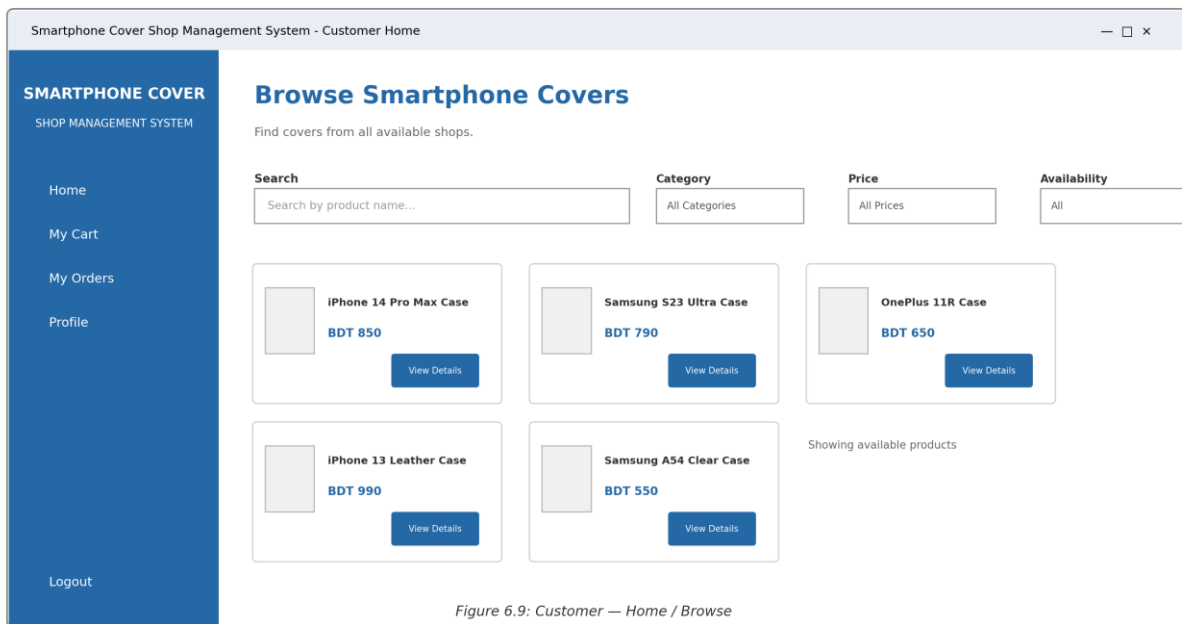


Figure 6.9: Customer — Home / Browse

The Customer can browse smartphone covers, search by product name and use Category, Price and Availability filters.

6.10 Customer — Product Details + Reviews

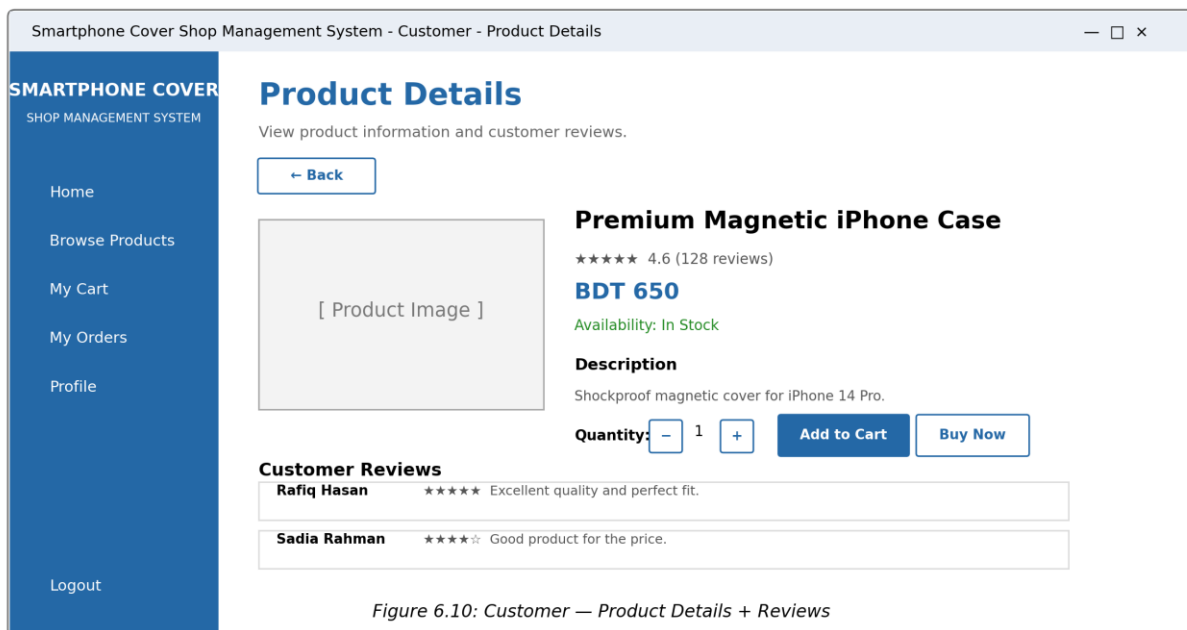


Figure 6.10: Customer — Product Details + Reviews

The Customer can view product details, availability, price, ratings and reviews before adding the product to the cart.

6.11 Customer — Cart

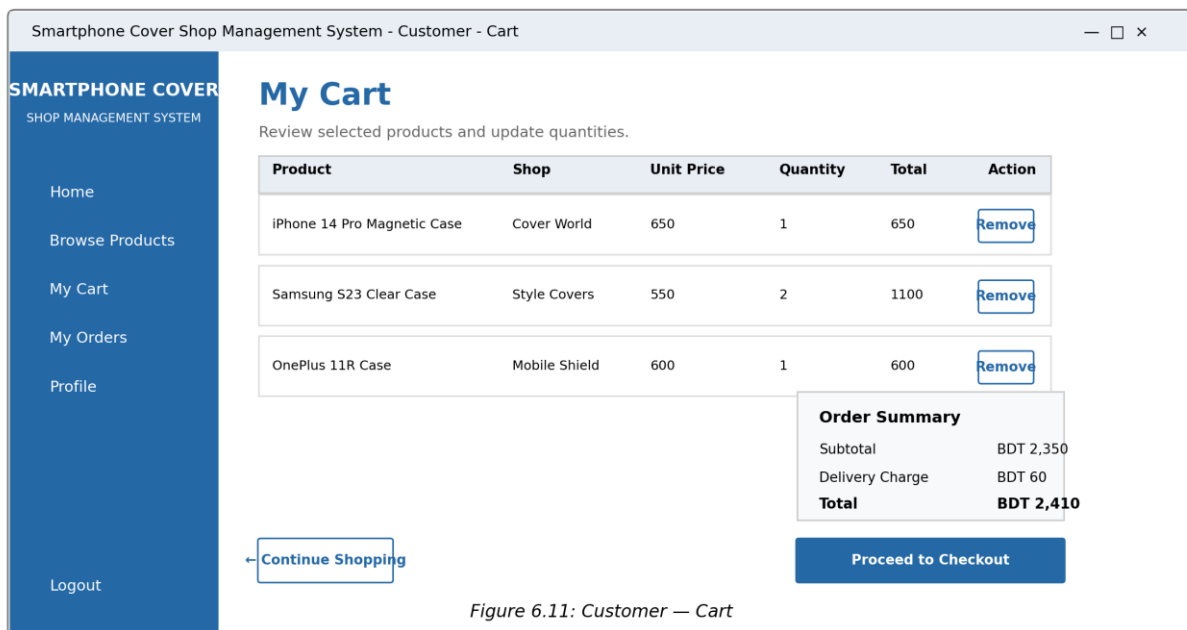


Figure 6.11: Customer — Cart

Figure 6.11: Customer — Cart

The Cart form allows the Customer to review selected products, change quantities, remove items and proceed to checkout.

6.12 Customer — Checkout / Invoice

Smartphone Cover Shop Management System - Customer - Checkout / Invoice

SMARTPHONE COVER SHOP MANAGEMENT SYSTEM

Checkout / Invoice

Complete the order and view the generated invoice.

Delivery Information

Full Name
Rafiq Hasan

Phone
01712-345678

Address
House 12, Road 5, Dhaka

Payment Method
Cash on Delivery

Invoice Summary

Product Total	BDT 2,350
Delivery	BDT 60
Grand Total	BDT 2,410

[Place Order](#) [Print Invoice](#)

Logout

Figure 6.12: Customer — Checkout / Invoice

Figure 6.12: Customer — Checkout / Invoice

The Checkout form captures delivery and payment information, displays the invoice summary and allows the Customer to place the order.

Chapter 7 — Work Distribution

The project work was distributed among the four group members according to the major components of the system and report. Nahiyen Sajid Saad was assigned the larger share of database/query and documentation-related work, while the remaining tasks were distributed among the other members.

Member Name	Student ID	Exact Contribution
Ahamed Mashuf Tasin	22-46073-1	Functional Requirements & User Stories; System Features and Role Analysis; C# application structure; Customer-side functionality
Zarif al afraim	21-44568-1	Database Design; SQL Schema Diagram; ER Diagram; 3NF Normalization; SQL Server database implementation
Nahiyen Sajid Saad	24-58497-2	Case Study; User Study; SQL Query List / Chapter 5 SQL Queries; UI Diagrams / Navigation Design; database-related analysis and documentation
Shehan Fazle Rabbi	22-48155-2	User Interface Design and Forms; Admin and Super Admin functionality; Testing and debugging; Documentation and final report compilation

Chapter 8 — Conclusion & Future Work

8.1 Conclusion

The Smartphone Cover Shop Management System provides a simple and organized solution for managing a multi-shop smartphone cover marketplace. The system connects customers with shop owners through a centralized platform managed by a Super Admin.

The system defines three main roles: Super Admin, Admin (Shop Owner), and Customer. The Super Admin manages the overall platform, shop owners manage their own products, stock and sales, and customers can browse products, add items to their cart, place orders and provide reviews.

The database is designed with appropriate primary and foreign keys and normalized to Third Normal Form (3NF) to reduce data redundancy and maintain data consistency. The SQL queries support important functions such as product searching and filtering, cart management, checkout, sales reporting, low-stock detection, reviews and offers.

The proposed Windows Forms interface provides a simple navigation structure with role-based access. Overall, the system provides a practical foundation for managing smartphone cover sales while reducing manual work and improving data organization.

8.2 Future Work

In the next stage of development, the designed system will be implemented as a C# Windows Forms application with database integration.

Implement the three role-based dashboards.

Connect all forms to the SQL database.

Implement product CRUD and inventory management.

Implement cart and checkout functionality.

Implement payment and order processing.

Implement reviews and discount offers.

Add proper input validation and error handling.

Test the complete system with different users and transactions.

8.3 Expected Challenges

The main challenges may include maintaining correct role-based access, keeping shop-owner data isolated, maintaining database consistency during orders and stock updates, and ensuring that the different forms and SQL queries work together correctly.

Technology Stack & Planned Execution

Technology Stack

C# with .NET for application development.

Windows Forms for the desktop graphical user interface.

Microsoft SQL Server / SQL Server Express for relational database storage.

Visual Studio for development and Windows Forms design.

How to Run (Planned)

Open the project in Visual Studio.

Configure the SQL Server database connection.

Create the project database and tables using the database schema script.

Insert the required sample data.

Build and run the Windows Forms application.

Log in using an account and verify that the correct role-based dashboard is displayed.